

Chapter 5

Money and the national debt

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1 Introduction

An important aspect of economic coordination is the willingness and ability of individuals to make and receive “gifts” in a multilateral arrangement. This scenario is captured in the OLG model when bilateral credit arrangements result in an inefficient outcome (the dynamic inefficiency case studied in the previous chapter). We saw how a public pension scheme could be designed to improve economic efficiency. But there is also a way to improve efficiency without the use of taxes and transfers as redistributive tools. The government could instead introduce a nominal security like currency (legal tender). Or the government could introduce bonds that are purchased by the banking sector (using deposit liabilities). This latter case corresponds to “full-reserve” banking; see Diagram 5.1.

Diagram 5.1

Fed		Banks		Households	
...Assets...	Liabilities	...Assets...	Liabilities	...Assets...	Liabilities
USTs	Reserves	Reserves	Deposits	Deposits	Wealth

So, the idea I have in mind here is that the government issues a quantity of zero-interest treasury securities D that are purchased by the Federal Reserve for M dollars of zero-interest reserves credited to the government’s TGA (Treasury General Account). The Treasury then transfers M dollars

to the initial old, who see money appearing in their checking accounts (there is a corresponding flow of reserves out of the TGA into the banking system). At this point, the household sector has financial assets in the form of bank deposits (held entirely by the initial old). These deposits (which are non-etized USTs) constitute part of household financial wealth. But one might alternatively assume that the government creates cash M , which it injects as a transfer to individuals. In this latter case, we can ignore the balance sheets for the Fed and Banks in Diagram 5.1, and replace “deposits” with “cash” in the balance sheet for Households. To begin, our analysis will assume a fixed supply of zero-interest money (which may alternatively be viewed as a perpetual stock of outstanding government debt).

2 Model

Consider our basic OLG model with 2-period-lived individuals. The population of young individuals is given by N_t and population grows at the constant gross rate n , so that $N_t = nN_{t-1}$. For simplicity, assume that $(y_1, y_2) = (y, 0)$. In what follows, I restrict attention to stationary allocations, so that $\{c_1(t), c_2(t+1)\} = \{c_1, c_2\}$ for all t . The representative young generation has preferences $U(c_1, c_2)$. Because $MRS(y, 0) < n$, this economy exhibits a dynamic inefficiency.

Let p_t denote the price of output measured in dollars. As in the previous chapter, we can think of a dollar as being a unit of account, even if no actual dollars exist (either physically or as book-entry objects). In this chapter, however, dollars actually exist (again, either physically or as book-entry objects). In particular, there is a fixed supply of M dollars (created out of D dollars of government securities). These dollars are tradeable. That is, one dollar can buy $1/p_t$ units of output at date t . Equivalently, one unit of output at date t can be purchased with p_t dollars.¹

¹Again, to emphasize the point, I mean “purchase with p_t dollars.” I do not mean “purchase with goods and services worth p_t dollars.” The distinction here is between a unit of account (the latter case) and a medium of exchange (the former case).

2.1 Individual choices

A young person at date t produces y units of output worth $p_t y$ dollars. If this person purchases c_1 units of output, it will cost $p_t c_1$ dollars. Because $y_1 > 0$ and $y_2 = 0$, the young will want to save. We know from the previous chapter that if the young are homogeneous, a private credit market serves no purpose. There is no point of saving by lending goods to another young person with the latter promising to repay goods (with interest) in the following period. This is because the young are all the same (homogeneous). Lending goods to old makes no sense either since the old will not be around to repay the loan in any future period. However, if the old have money, the young might be interested in *selling* some of their output for money,

$$m_t = p_t y - p_t c_1 \quad (1)$$

Why might this make sense? Well, ask yourself why you are willing to sell your labor for money. The answer (I think) is because the money you acquire can be used some time in the future to buy the things you want. Our model people are no different from you in this regard. A person who acquires m_t dollars in period t will be able to spend m_t dollars in period $t+1$ (remember, I am assuming money bears no interest here); i.e.,

$$p_{t+1} c_2 = m_t \quad (2)$$

Combine (1) and (2) to form $p_{t+1} c_2 = p_t y - p_t c_1$ and rearrange to form,

$$c_2 = \Pi^{-1} y - \Pi^{-1} c_1 \quad (3)$$

where $\Pi \equiv p_{t+1}/p_t$ is the (gross) rate of inflation.² Equation (3) is the budget line. Note how (3) resembles the budget line derived in Chapter 4 under a private credit market with $(y_1, y_2) = (y, 0)$; namely, $c_2 = ry - rc_1$. Recall that in a private credit market, r represented the real rate of interest on privately-issued securities (representing dollar-denominated claims against output). Evidently, we can think of Π^{-1} as the real rate of return on money.

²In keeping with my assumption of stationarity, I assume that the rate of inflation is constant over time. This need not be the case. Non-stationary equilibria are also possible, but I relegate this topic to the appendix.

Exercise 1 Draw the budget line (3) on a diagram. What is the slope of the budget line? Note that c_2 is a good that can only be purchased using money, while c_1 is a good that can be purchased without money (in reality, leisure is a good example of such a good). Explain how the inflation rate can be thought of as influencing the relative price between cash and non-cash goods.

We can follow the protocol developed in earlier chapters. A young person chooses (c_1, c_2) to maximize $U(c_1, c_2)$ subject to the budget constraint (3). The solution is a pair of demand functions (c_1^D, c_2^D) that satisfy the following two conditions:

$$MRS(c_1^D, c_2^D) = \Pi^{-1} \quad (4)$$

$$c_2^D = \Pi^{-1}y - \Pi^{-1}c_1^D \quad (5)$$

The desired saving function can then be derived as $s^D = y - c_1^D$. Note that because money is the only security in this economy, and because only the young wish to accumulate and hold money over time, s^D also represents the demand for real money balances.³

Exercise 2 For $MRS = (c_2/c_1)^{1/2}$, use (4)-(5) to derive the demand for real money balances $s^D(y, \Pi)$. How does the demand for real money balances depend on the expected rate of inflation? Explain. Hint: think of the cash vs. non-cash goods distinction made above.

2.2 Determining the price-level and inflation

In this environment, the young want money and are willing to sell goods to get it. But who has the money? In this model, the entire money supply M is held by the old at the start of any date t . To compute national saving in this economy, we have to take into account the fact that the old are now in a position to *dissave* (i.e., spend their accumulated savings). How much do the old, in aggregate, wish to dissave? Because the old in this model have no motivation to carry wealth beyond the grave (there is no bequest motive), they will want to spend all of their money. In equilibrium, we know that desired national saving must be equal to zero, i.e.,

$$N_t s^D(y, \Pi) - \frac{M}{p_t} = 0 \quad (6)$$

³The demand for nominal money balances is given by $m_t^D = p_t s^D$.

We can use condition (6) to solve for the equilibrium inflation rate, Π . To do so, note that because (6) holds at all dates t , it must hold at date $t + 1$,

$$N_{t+1}s^D(y, \Pi) - \frac{M}{p_{t+1}} = 0 \quad (7)$$

Conditions (6) and (7) together imply that

$$\Pi^* = 1/n \quad (8)$$

For a fixed money supply M , this model predicts that the equilibrium inflation rate is inversely related to RGDP growth (here, equivalent to population growth). Think of n as representing the growth rate of the demand for real money balances. If the demand for money grows over time and the supply of money is fixed, then the purchasing power of money is likely to increase over time—i.e., the economy will experience a *deflation*.

Proposition 5.1 *The level of money (government debt) M does not influence the long-run inflation rate.*

With the equilibrium inflation rate determined in this manner, we can use condition (6) to determine the equilibrium price-level,

$$p_t^* = \frac{M}{N_t s^D(y, \Pi^*)} \quad (9)$$

for $t = 1, 2, \dots, \infty$. Condition (9) is consistent with the so-called *Quantity Theory of Money*, which asserts that the price-level is proportional to the stock of money.

Exercise 3 *The real GDP, which is given by $N_t y$, may either grow or contract with population. Explain why the nominal GDP, $Y_t = p_t N_t y$ stays constant over time in this economy whether or not the real GDP is growing or contracting.*

2.3 Is the monetary equilibrium efficient?

Using the analysis from the previous chapter, we know that the competitive equilibrium of a private credit economy is inefficient for this economy. How does the competitive equilibrium of this monetary economy compare?

Let us denote the equilibrium allocation as (c_1^e, c_2^e) . In equilibrium, individuals are optimizing (doing the best they can, according to their preferences, and subject to their constraints). This means that (c_1^e, c_2^e) must satisfy conditions (4)-(5); i.e.,

$$MRS(c_1^e, c_2^e) = \Pi^{-1} \quad (10)$$

$$c_2^e = \Pi^{-1}y - \Pi^{-1}c_1^e \quad (11)$$

Next, we know that market-clearing (6) at each and every date implies an equilibrium inflation rate (8). Combining (8) with (10)-(11), we derive,

$$MRS(c_1^e, c_2^e) = n \quad (12)$$

$$c_2^e = ny - nc_1^e \quad (13)$$

Since conditions (12)-(13) are exactly those that characterize the Golden rule allocation, we have the rather remarkable result that $(c_1^e, c_2^e) = (c_1^*, c_2^*)$. That is, the competitive monetary equilibrium implements the Golden rule allocation, with an equilibrium real rate of return on money equal to the population growth rate. Moreover, note that implementation requires no fiscal policy (apart from the initial inject of government debt into the economy). This is in contrast to the social security system studied in the previous chapter.

3 The national debt

The nominal national debt in this economy is given by $D = M$. The idea here is that the U.S. Treasury issues securities D that are then “monetized” in some manner, either by the central bank, or by the private bank sector. But we can also think of the U.S. Treasury itself as issuing its securities in a form suitable for making payments. I discussed in Chapter 1 how the U.S. Treasury issues currency in the form of small-denomination coins. But from 1862-1971, the U.S. Treasury also issued small-denomination paper bills called *U.S. Notes* (also known as *Greenbacks*).⁴ Remember, the U.S. Federal Reserve was established only in 1913, so U.S. Treasury money preceded Federal Reserve money. Figure 5.1 displays a five-dollar Federal Reserve note side by side with a five-dollar U.S. Note. Can you spot the difference?

⁴See: https://en.wikipedia.org/wiki/United_States_Note

Why is it customary to label the top note in Figure 5.1 “money” and the bottom note “debt?” We will come back to this question later on, but for now (and perhaps even later on), the answer should not be obvious. It is true that at various stages the U.S. Note was redeemable for specie (gold or silver coin). But the same is true of Federal Reserve notes. In any case, both notes today are not redeemable for anything and both notes constitute legal tender.

So let us imagine then that the national debt M consists of notes like the ones displayed in Figure 5.1. Or it might consist of zero-interest checking accounts in the banking system (with private bank deposit liabilities matched by a corresponding value of government securities held as assets). Any of these interpretations is fine for the discussion that follows.

Figure 5.1
Federal Reserve Note



U.S. Note



If we take a snapshot of our economy at the beginning of any date t , we will see that the national debt is held by the old as a financial asset. The national debt is equal to private sector financial wealth. National financial wealth is still equal to zero in this economy. A commonly cited statistic is the debt-to-GDP ratio. If we let D_t denote government debt and let Y_t denote

nominal GDP (NGDP), then the debt-to-GDP ratio is given by,

$$\left(\frac{D_t}{Y_t}\right) = \left(\frac{M}{p_t N_t y}\right) \quad (14)$$

where $N_t y$ is the real GDP (RGDP). If we combine (9) with (14), we derive the equilibrium debt-to-GDP ratio,

$$\left(\frac{D_t}{Y_t}\right) = \left[\frac{s^D(y, \Pi^*)}{y}\right] \quad (15)$$

Proposition 5.2 *The long-run debt-to-GDP ratio does not depend on the level of nominal debt.*

I use the “long-run” qualifier in the proposition above because our analysis is restricted to studying stationary equilibria. Changes in the level of nominal debt D_t are likely to affect the D_t/Y_t ratio in the short-run (we will consider such a case below).

The intuition for Proposition 5.2 is as follows. First note from Proposition 5.1 that the *level* of debt does not influence the long-run inflation *rate*. Second, note that it is *assumed* here that the level of debt does not influence real personal income y . Given the former result and the latter assumption, the model produces a “long-run debt neutrality” result. A larger nominal debt D_t ultimately manifests itself as a proportionately larger price-level—see equation (9). That is, both D_t and Y_t (the NGDP) increase in the same proportion.

The idea that the outstanding stock of debt ultimately manifests itself as higher NGDP (via a higher price-level) is not inconsistent with the experiences in many countries. In Figure 5.2, I plot the national debt and NGDP for Canada. The ratio varies over time, but they do seem to grow in

sympathy with each other.

Figure 5.2

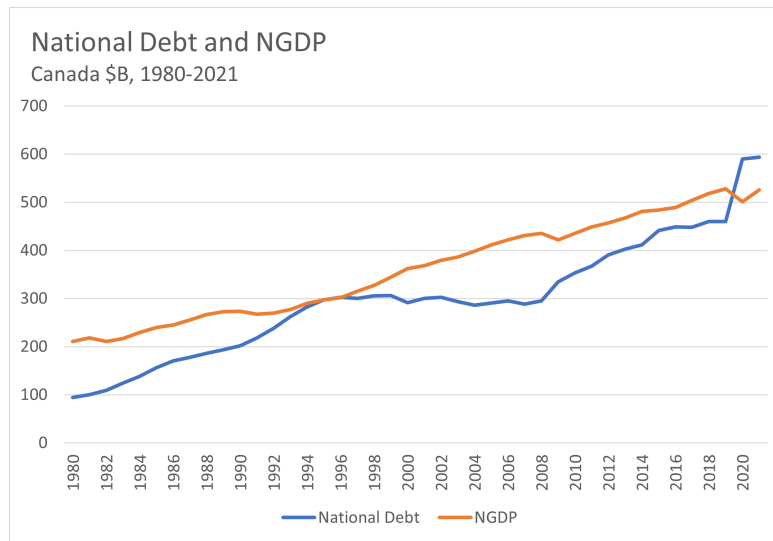


Figure 5.3 plots the debt-to-GDP ratio for a few countries.

Figure 5.3

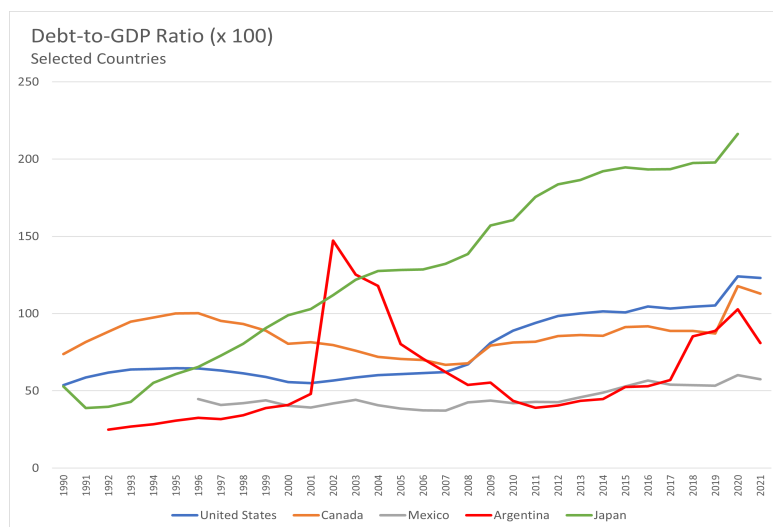


Figure 5.3 shows how the debt-to-GDP varies across countries and how it

can vary within a country over time. In Argentina, there are two large spikes, but the long-run ratio is lower than in many other countries. Mexico's ratio is relatively low and stable. Japan's ratio has been climbing for decades.

Japan's case is instructive. It seems unlikely that the debt-to-GDP ratio in Japan has been increasing simply because the government of Japan is issuing more debt. After all, we see from Figure 5.2 that the Canadian government is also issuing more debt. Part of the answer has to lie in the *demand* for government debt. The demand for debt in our theory is related to the desired saving function, $s^D(y, \Pi)$. To understand why some countries are able to sustain relatively high debt-to-GDP ratios, we need to understand why these countries sovereign debt is so highly valued.

3.1 Interest-bearing national debt

Apart from currency, most of the national debt bears interest. How does “interest-bearing money” affect our analysis? Let M_t denote the outstanding supply of debt at date t and let R_t denote the gross nominal interest rate paid on this debt at date $t + 1$. Then the amount of interest owed by the government at date t is given by $(R_{t-1} - 1)M_{t-1}$. If M_t consisted entirely of currency, then $R_t = 1$ for all t and the interest expense of the debt would be zero. But suppose $R_t > 1$. How is this interest expense financed?⁵ In general, this can be done in one of two ways. First, through tax revenue T_t . Second, through the issuance of new debt (money), $M_t - M_{t-1}$. We can write the government budget constraint as,

$$(R_{t-1} - 1)M_{t-1} = T_t + M_t - M_{t-1} \quad (16)$$

For simplicity, let us consider a constant nominal interest rate, $R_t = R$. As well, assume that the nominal debt grows (or shrinks) at some constant rate, μ , so that $M_t = \mu M_{t-1}$. Note that this latter expression implies $M_{t-1} = M_t/\mu$ which, if we combine with (16), allows us to write the government budget constraint as,

$$T_t = \left[\frac{R}{\mu} - 1 \right] M_t \quad (17)$$

⁵If $R_t < 1$, then the nominal interest rate is negative and people are paying the government for the privilege of storing nominal wealth in the form of government securities.

The left-hand-side of (17) represents the government's primary budget surplus.⁶ Condition (17) tells us that if the government pays interest $R > \mu$ on the debt, then it must run a primary surplus to finance the interest expense. If $R < \mu$, then the government can run a primary budget deficit. If $R = \mu$, then the government is using new money to finance its entire interest expense, so that the primary surplus/deficit is zero.

Let us suppose that taxes are paid by the old (the people with financial wealth in our economy). To calculate the tax paid by each old person, divide both sides of (17) by N_{t-1} . Also, to convert the measurement from dollars to goods, divide both sides of (17) by the price-level, p_t , to derive,

$$\left(\frac{T_t}{N_{t-1}p_t} \right) = \left[\frac{R}{\mu} - 1 \right] \left(\frac{M_t}{N_{t-1}p_t} \right) \quad (18)$$

Define $\tau \equiv [T_t/(N_{t-1}p_t)]$ and use $N_{t-1} = N_t/n$ to rewrite (18) as,

$$\tau = \left[\frac{R}{\mu} - 1 \right] n \left(\frac{M_t}{N_t p_t} \right) \quad (19)$$

The restriction to stationary outcomes implies that the real tax paid by each old person remains constant over time even though the total real tax grows with population and the total nominal tax paid grows owing to population growth and inflation.

Now let us ask how interest-bearing debt might be expected to influence economic activity. Let us reconsider the decisions people make when there is interest-bearing debt and taxation. Recall that only the old are assumed to pay the tax here. In this case, the budget constraint for a young person remains unchanged from (1), $m_t = p_t y - p_t c_1$. However, the budget constraint for the old (2) now becomes $p_{t+1} c_2 = R m_t - T_{t+1}/N_t$. If we combine this latter

⁶In general, the primary surplus is given by $T_t - G_t$, where G_t represents government spending. The analysis here sets $G_t = 0$.

expression with (1), we get⁷

$$c_2 = (R/\Pi)y - (R/\Pi)c_1 - \tau \quad (20)$$

Comparing (20) with (3), we see that the lump-sum tax lowers the intercept of the budget line while $R > 1$ increases its slope.⁸ The conditions for individual optimality are now given by,

$$MRS(c_1^D, c_2^D) = R/\Pi \quad (21)$$

$$c_2^D = (R/\Pi)y - (R/\Pi)c_1^D - \tau \quad (22)$$

Conditions (21)-(22) characterize the demand functions $c_1^D(y, R/\Pi, \tau)$ and $c_2^D(y, R/\Pi, \tau)$. Using the definition of saving, $s = y - c_1$, we can again derive the desired saving schedule for a young person $s^D(y, R/\Pi, \tau)$.

Exercise 4 *Explain how desired saving by the young depends on the expected future transfer, τ .*

We know that national saving in this economy must be equal to zero. Our theory assumes that prices and quantities will adjust to ensure that desired national saving is equal to zero in equilibrium. Because the debt is now potentially changing over time, condition (6) is now written as,

$$N_t s^D(y, R/\Pi, \tau) - \frac{M_t}{p_t} = 0 \quad (23)$$

for all t . Because $s^D(y, R/\Pi, \tau)$ remains constant over time, condition (23) implies that the equilibrium inflation rate is given by,

$$\Pi^* = \frac{\mu}{n} \quad (24)$$

⁷The steps are as follows:

$$\begin{aligned} p_{t+1}c_2 &= Rp_t y - Rp_t c_1 - T_{t+1}/N_t \\ c_2 &= (R/\Pi)y - (R/\Pi)c_1 - \frac{T_{t+1}}{p_{t+1}N_t} \\ c_2 &= (R/\Pi)y - (R/\Pi)c_1 - \tau \end{aligned}$$

⁸The opposite would hold if $T_t < 0$ and $R < 1$.

Exercise 5 *Explain the economic intuition behind (24). Hint: think of the numerator as expressing “supply” and the denominator as expressing “demand.”*

Let us now characterize the competitive equilibrium allocation (c_1^e, c_2^e) . Combining (21) and (24), we know

$$MRS(c_1^e, c_2^e) = Rn/\mu \quad (25)$$

Notice how condition (25) corresponds to our earlier condition (12) when $R/\mu = 1$. This latter condition obviously holds when $R = \mu = 1$; that is, when we are considering non-interest-bearing debt and a constant supply of debt. The effect of $R/\mu > 1$ is to make the equilibrium budget constraint steeper.

Exercise 6 *Explain why a steeper budget line $(R/\mu) > 1$ increases the price of c_2 (future consumption) relative to c_1 (current consumption) for young individuals.*

Condition (25) tells us that at the equilibrium allocation (c_1^e, c_2^e) , the slope of the budget line will be steeper than the slope of the feasible line. But where will the new allocation line lie on the equilibrium budget line,

$$c_2^e = (R/\mu)ny - (R/\mu)nc_1^e - \tau^e \quad (26)$$

To answer this question, consider the following calculations. First, from the market-clearing condition (23), we have

$$s^e = y - c_1^e = \left(\frac{M_t}{N_t p_t} \right) \quad (27)$$

Next, combine (27) with the government budget constraint (19) to derive,

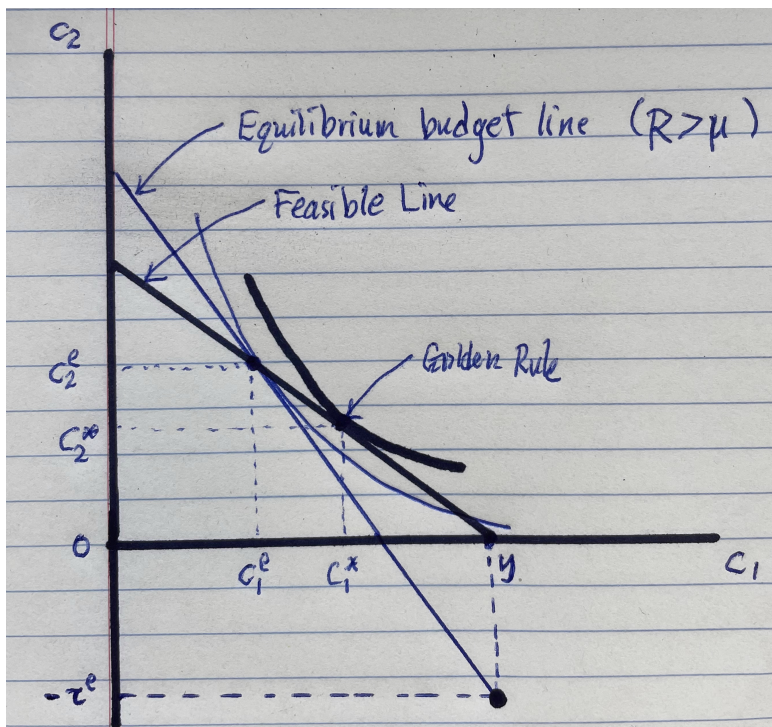
$$\tau^e = [R/\mu - 1] n (y - c_1^e) \quad (28)$$

Finally, combine (28) with (26) to derive,

$$\begin{aligned} c_2^e &= (R/\mu)ny - (R/\mu)nc_1^e - \tau^e \\ c_2^e &= (R/\mu)ny - (R/\mu)nc_1^e - [R/\mu - 1] n (y - c_1^e) \\ c_2^e &= ny - nc_1^e \end{aligned} \quad (29)$$

Condition (29) tells us that the new allocation lies on the feasible line. It follows that if $(R/\mu) > 1$, then $c_1^e < c_1^*$ and $c_2^e > c_2^*$. The new equilibrium is depicted in Figure 5.3.

Figure 5.3



Exercise 7 Characterize the equilibrium allocation (c_1^e, c_2^e) when $R/\mu < 1$. (Note: in this case, $\tau^e < 0$).

3.2 Is the national debt a burden for future generations?

It is commonly asserted that the national debt is a burden for future generations.⁹ The logic is simple to understand from an individual's perspective: if you accumulate debt when young, you will have to repay it when you are old.

⁹See, for example, <https://www.concordcoalition.org/issue-brief/passing-the-buck-how-the-national-debt-burdens-future-generations/>

The short run gain is offset by a long run pain. Should the same principle not apply to governments?

The simple (and compelling) “household analogy” of government debt, however, is misleading at best.¹⁰ We can ask current and future generations in our model how they feel about the debt, say, relative to a world without government debt (as in the previous chapter). The existence of government debt in the model developed above actually improves the economic welfare of everyone in the economy (relative to a world without government debt). While the young have to work to acquire financial assets, they plan to use these assets later on in their life cycle. The national debt serves as a type of exchange medium in this economy.

Can there be too much government debt in the economy? One way to answer this question is to consider two model economies that are identical in every respect except for the level of nominal government debt, M . According to our model (see condition 9), the two economies would differ only in their respective price-levels. While the cost-of-living would be higher in the high nominal debt economy, the citizens of this economy would have correspondingly more financial wealth. Real material living standards are invariant to the level of nominal debt in this economy.

What about the interest expense of the debt? While it is true that a higher interest rate increases the cost of financing the debt, we need to keep in mind that interest paid equals interest received by whoever is holding that debt. In our model, a higher interest rate may imply a higher future tax burden for individuals, but this burden is offset to some extent by higher future interest income. It is not immediately clear what the intergenerational consequences of interest-bearing debt are.

These types of results should lead us to think more deeply about what it is that should concern us about “excessive government debt.” We should keep in mind, however, that our model leaves out several real world considerations that may be important for understanding this question.

¹⁰See, for example, <https://www.stlouisfed.org/publications/regional-economist/fourth-quarter-2020/does-national-debt-matter>

4 Appendix 1: Aggregate demand deficiency

To this point, I have presented an endowment economy. This is just a fancy way of saying that the model does not explain where production y comes from—it is simply endowed to us like manna from heaven.

It is an easy manner to generalize our model in a way that permits output to be determined in some manner. The neoclassical approach is to assume that production depends on factor inputs, like labor. Let us now assume that the young are endowed with unit of time and let $0 \leq n \leq 1$ denote the time that the young spend at work. The remaining time $1 - n$ is spent in activities that generate value not counted in market production (RGDP). Assume that $y = \alpha n$, where $\alpha > 1$ is a productivity parameter. Suppose individuals make a cost-benefit analysis that leads them to choose some level of employment, n^* . Then their real income is given by $y^* = \alpha n^*$ and the RGDP of this economy is given by $Y_t^* = N_t y^*$. Everything we learned above will continue to hold with $y = y^*$.

The neoclassical model of production is sometimes criticized for assuming “full employment.” Or, at least, it is criticized for assuming conditions that imply that the economy coordinates its resources efficiently. It is not entirely clear that such an assumption is always warranted. In general, we cannot rule out cases in which the economy (through whatever mechanisms) fails to coordinate resources efficiently. This is what Keynes (1936) meant by a *General Theory*. The Great Depression was, in a view shared widely by others, an episode where the economy was clearly operating at under full capacity (with the implication that the situation was highly inefficient).

One way to model this idea is to assume that y is “demand determined,” at least, up to some capacity constraint, so that $y \leq \bar{y}$. The neoclassical model specifies conditions under which $y = \bar{y}$. But what if, for some reason, the economy coordinates on a sub-optimal level of employment/production, $y < \bar{y}$. Suppose, for example, that the young do not value what they produce—they need to exchange it for the output produced by other young people. If everyone works to capacity and swaps goods, everyone earn $\bar{y}$ (the neoclassical outcome). But what if people only expect to sell $y < \bar{y}$ units of output? Then they have no reason to work harder than $n = y/\alpha$. And if everyone feels the same way, it is a Nash equilibrium to work $n = y/\alpha$ hours. The economy coordinates on an “underemployment equilibrium.”

Because the analysis above is performed assuming a given level of y , it continues to hold in this scenario. Specifically, conditions (4)-(5) still describe a young person's optimal consumption-saving choice, conditional on a level of personal income, y . This gives rise to the same desired saving function $s^D(y, \Pi)$. And once again, we can assume that desired national saving must be equal to zero in equilibrium; i.e., from condition (6),

$$N_t s^D(y, \Pi) = \frac{M}{p_t} \quad (30)$$

Again, it follows from (30) that $\Pi^* = 1/n$. The only thing that differs from our earlier analysis is that y is now determined by a self-fulfilling prophecy ($y < \bar{y}$) instead of economic fundamentals ($y = \bar{y}$).

An economy that finds itself in a “low-level” equilibrium $y < \bar{y}$ can be thought of experiencing “deficient aggregate demand.” Firms are not hiring all the workers they might because they do not expect the demand for their product to materialize. If all firms think the same way, a low level of output and employment may be the result. In this case, there may be role for policy to “stimulate” aggregate demand. One way in which this could be done here is by increasing the money supply M in the hands of the public. Then, assuming that the price-level p_t does not rise in proportion to the change in M (in the extreme case, assume that the price-level remains fixed as long as demand is deficient), then condition (30) implies that an increase in demand (M/p_t) will elicit an increase in output y ; at least, as long as $y < \bar{y}$. If M is increased again after the economy has achieved “full employment” ($y = \bar{y}$), then the result will be an increase in p_t (a temporary inflation).

5 Appendix 2: Short-run money non-neutrality

Equation (9) tells us that money is “neutral” in the sense that a change in the money supply M does not have any implications for the way resources are allocated in the economy. A larger M simply means a larger p_t ; at least, assuming that the economy is operating at “full employment.”

Again, this neutrality proposition is best thought of as a statement of what is likely to happen in the “long run.” As it turns out, money is not neutral in the short-run in this model economy (even if the economy is operating at full employment).

Consider the following experiment. Suppose we are initially in a stationary equilibrium with money supply M_L . Suppose that the government suddenly (unexpectedly) creates $M_H - M_L > 0$ new dollars that it injects into the economy as a lump-sum transfer. That is, everyone alive at date $t = t_0$ (young and old) receive an equal share of the new money created. The money supply after the transfer is M_H and is expected to remain fixed forever. What are the short and long-run consequences of this “helicopter drop” of money?

We already know from (9) and Proposition 1 that prior to date t_0 the price-level is given by

$$p_t^L = \frac{M^L}{N_t s^D(y, 1/n)}$$

and that after a sufficiently long period of time, the price-level is given by

$$p_t^H = \frac{M^H}{N_t s^D(y, 1/n)}$$

But what happens in the short-run?

The answer is fairly easy to deduce. Prior to date t_0 , the young enter the economy with zero money balances. They must go to work (producing and selling output, y) to accumulate financial wealth. This proposed helicopter drop of money gives the young at date t_0 a gift of money that, for a given price-level, increases their wealth. The young will want to spend some of this added wealth today and save some of it for the future. The old, of course, simply want to spend their entire financial wealth, now increased with the transfer as well. Together, the young and the old will bid up the price of output, the supply of which remains fixed at $N_{t_0}y$. So, we know that p_{t_0} will increase relative to where it would have been had the transfer not taken place. But more importantly, we know that the young at date t_0 must also enjoy a greater level of consumption $c_1(t_0) > c_1$ than they would have enjoyed had the transfer not taken place. Since aggregate output is fixed, this extra consumption can only take place at the expense of the old at date t_0 ; i.e., $c_2(t_0) < c_2$. Because the money transfer affects resource allocation at date t_0 , money is not neutral in the short run.¹¹ For dates $t > t_0$, the economy transitions back to the stationary equilibrium, albeit on a higher price-level path.

¹¹Note that this is true despite the fact that prices are assumed to be perfectly flexible in this thought experiment.

In general, the economic effects of a money transfer will depend on the targeted recipients. In this model, if the entire transfer is targeted to the old at date t_0 , one can show that the money transfer is neutral in the short and long run. The old in this economy have a very high MPC (marginal propensity to consume). If their financial wealth is increased by some factor, they will simply end up bidding up the price of goods and services proportionately.

Suppose, on the other hand, that the money transfer is targeted to the young at date t_0 . This is a case where “new money” competes against “old money” for scarce output. As the price-level is bid up, the purchasing power of “old money” declines. While the purchasing power of “new money” also declines, at least the young have more money to compensate for the higher cost of living. And so, the young will succeed in increasing their consumption share of the RGDP at the expense of the old.